

82. Required number

= H.C.F. of $(1356 - 12)$, $(1868 - 12)$ and $(2764 - 12)$
 = H.C.F. of 1344, 1856 and 2752 = 64.

83. Required number

= H.C.F. of $(3026 - 11)$ and $(5053 - 13)$
 = H.C.F. of 3015 and 5040 = 45.

84. Required number

= H.C.F. of $(964 - 41)$, $(1238 - 31)$ and $(1400 - 51)$
 = H.C.F. of 923, 1207 and 1349 = 71.

85. L.C.M. of 8, 16, 40 and 80 = 80.

$$\frac{7}{8} = \frac{70}{80}; \frac{13}{16} = \frac{65}{80}; \frac{31}{40} = \frac{62}{80}.$$

Since, $\frac{70}{80} > \frac{65}{80} > \frac{62}{80}$, so $\frac{7}{8} > \frac{13}{16} > \frac{31}{40}$. So, $\frac{7}{8}$ is the largest.

86. Required number

= L.C.M. of 1, 2, 3, 4, 5, 6, 7, 8, 9
 = $2 \times 2 \times 3 \times 5 \times 7 \times 2 \times 3 = 2520$.

$$\begin{array}{r} 2 \overline{) 1 - 2 - 3 - 4 - 5 - 6 - 7 - 8 - 9} \\ 2 \overline{) 1 - 1 - 3 - 2 - 5 - 3 - 7 - 4 - 9} \\ 3 \overline{) 1 - 1 - 3 - 1 - 5 - 3 - 7 - 2 - 9} \\ \hline 1 - 1 - 1 - 1 - 5 - 1 - 7 - 2 - 3 \end{array}$$

87. L.C.M. of 12, 18, 21, 30

= $2 \times 3 \times 2 \times 3 \times 7 \times 5$
 = 1260.

$\therefore$ Required number = $(1260 \div 2) = 630$.

88. Let the two numbers be a and b .

Then, $a + b = 45$... (i)

And, $a - b = 5$... (ii)

Adding (i) and (ii), we get: $2a = 50$ or $a = 25$.

Putting $a = 25$ in (i), we get: $b = 20$.

$\therefore$ L.C.M. = $5 \times 4 \times 5 = 100$.

89. Required fraction

= L.C.M. of $\frac{6}{7}, \frac{5}{14}, \frac{10}{21} = \frac{\text{L.C.M. of } 6, 5, 10}{\text{H.C.F. of } 7, 14, 21} = \frac{30}{7}$.

90. Least number of 5 digits is 10000. L.C.M. of 12, 15 and 18 is 180.

On dividing 10000 by 180, the remainder is 100.

$\therefore$ Required number = $10000 + (180 - 100) = 10080$.

91. Greatest number of 4 digits is 9999.

$$\begin{array}{r} 5 \overline{) 15 - 25 - 40 - 75} \\ 5 \overline{) 3 - 5 - 8 - 15} \\ 3 \overline{) 3 - 1 - 8 - 3} \\ \hline 1 - 1 - 8 - 1 \end{array}$$

L.C.M. of 15, 25, 40 and 75 = $5 \times 5 \times 3 \times 8 = 600$.

On dividing 9999 by 600, the remainder is 399.

$\therefore$ Required number = $(9999 - 399) = 9600$.

92. L.C.M. of 12, 18, 21 and 32 = $2 \times 2 \times 3 \times 3 \times 7 \times 8 = 2016$.

$$\begin{array}{r} 2 \overline{) 12 - 18 - 21 - 32} \\ 2 \overline{) 6 - 9 - 21 - 16} \\ 3 \overline{) 3 - 9 - 21 - 8} \\ \hline 1 - 3 - 7 - 8 \end{array}$$

So, the required number is a multiple of 2016 and lies between 4000 and 5000.

Hence, required number = 4032.

93. L.C.M. of 25, 50 and 75 = $5 \times 5 \times 2 \times 3 = 150$.

$$\begin{array}{r} 5 \overline{) 25 - 50 - 75} \\ 5 \overline{) 5 - 10 - 15} \\ \hline 1 - 2 - 3 \end{array}$$

On dividing 43582 by 150, the remainder is 82 and quotient is 290.

So, required number = $150 \times 291 = 43650$.

94. L.C.M. of 5, 6, 4 and 3 = 60. On dividing 2497 by 60, the remainder is 37.

$\therefore$ Number to be added = $(60 - 37) = 23$.

95. L.C.M. of 20, 28, 32, 35 = $2 \times 2 \times 5 \times 7 \times 8 = 1120$.

Required number = $(5834 - 1120) = 4714$.

$$\begin{array}{r} 2 \overline{) 20 - 28 - 32 - 35} \\ 2 \overline{) 10 - 14 - 16 - 35} \\ 5 \overline{) 5 - 7 - 8 - 35} \\ 7 \overline{) 1 - 7 - 8 - 7} \\ \hline 1 - 1 - 8 - 1 \end{array}$$

96. The least number divisible by 16, 20, 24

= L.C.M. of 16, 20, 24 = $240 = 2 \times 2 \times 2 \times 2 \times 3 \times 5$.

To make it a perfect square, it must be multiplied by 3×5 .

$\therefore$ Required number = $240 \times 3 \times 5 = 3600$.

97. Required number = (L.C.M. of 12, 16, 18, 21, 28) + 7

= $1008 + 7 = 1015$.

98. Required number = (L.C.M. of 24, 32, 36, 54) - 5

= $864 - 5 = 859$.

99. Required number = (L.C.M. of 12, 15, 20, 54) + 8

= $540 + 8 = 548$.

100. L.C.M. of 4, 5, 6, 7 = 420.

$\therefore$ Required number = $420 + 1 = 421$.

101. Greatest number of 3 digits is 999. L.C.M. of 6, 9 and 12 is 36.

On dividing 999 by 36, the remainder obtained is 27.

So, required number = $(999 - 27) + 3 = 975$.

102. Greatest number of 4 digits is 9999. L.C.M. of 4, 7 and 13 = 364.

On dividing 9999 by 364, the remainder obtained is 171.

$\therefore$ Greatest number of 4 digits divisible by 4, 7 and 13 = $(9999 - 171) = 9828$.

Hence, required number = $(9828 + 3) = 9831$.

103. Least number of 6 digits is 100000. L.C.M. of 4, 6, 10 and 15 = 60.

On dividing 100000 by 60, the remainder obtained is 40.

$\therefore$ Least number of 6 digits divisible by 4, 6, 10 and 15